Hertie School, MDS programme
Machine Learning, Spring 2025

FORECASTING WIND POWER GENERATION IN THE NETHERLANDS

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Problem Setting

Idea: Forecast energy generation volume from offshore and onshore wind power in the Netherlands

Task

Learn the mapping of the input features to the label

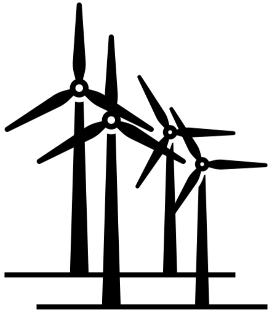
Experience

A set of feature-label pairs (training data)

Performance Measure

A measure for the quality of the predictions (MSE/RMSE)

The Data



- Scraped from Nationaal Energie Dashboard, KNMI, Jeroen (Bewus met Energie)
- Spans period from 01/01/2017 - 31/12/2024
- Data on wind power generation, meteorological factors and dynamic consumer prices

Advantages

1. Good data availability
2. Customizable granularity
3. Only three institutes required for all variables

Data Challenge 1 & 2: High Granularity + Time Series

Problems

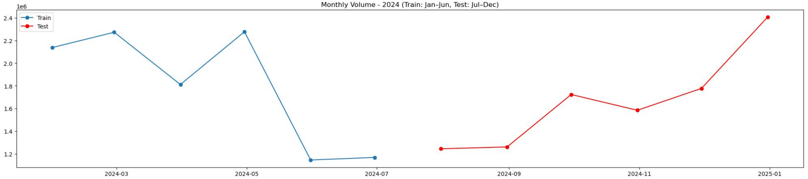
Exceptionally high temporal resolution:
70 128 data points
in each data set

Manually choosing test sets because randomisation does not work with time series

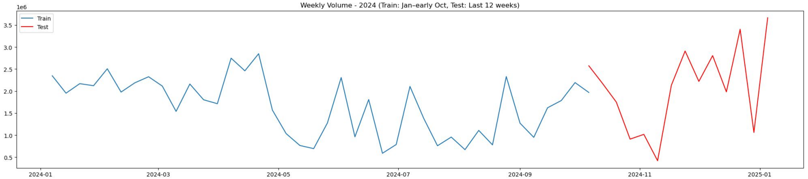
Selecting a train set that is feasible for fitting into models

Solution - temporal aggregation of data

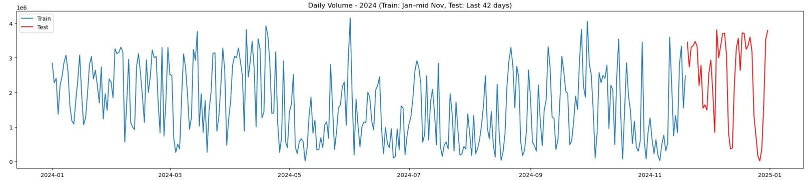
Monthly data:
96 data points
Test - **6 months**



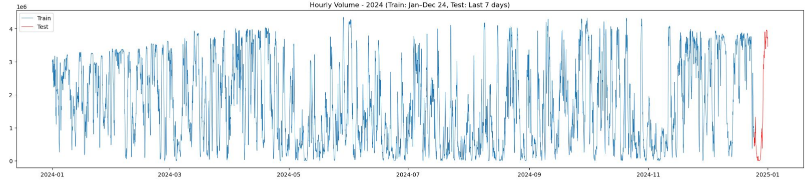
Weekly data:
418 data points
Test - **12 weeks**



Daily data:
2 922 data points
Test - **60 days**



Hourly data:
70 128 data points
Test - **168 hours** (7 days)



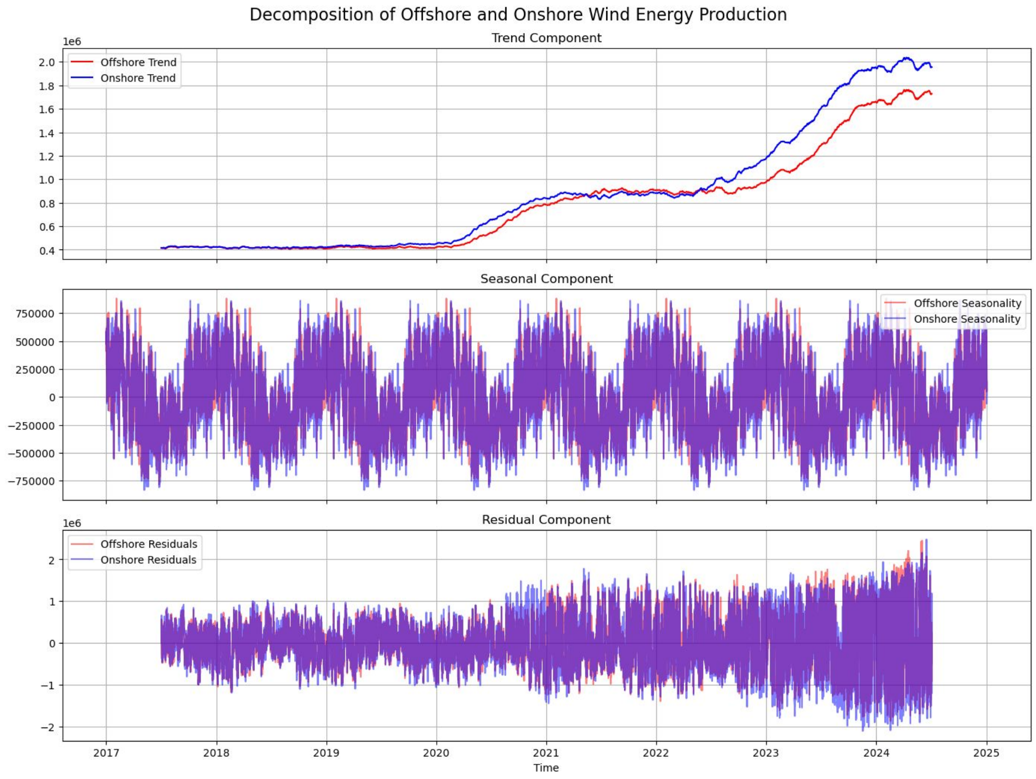
Data Challenge 2: Non-Stationarity

Problems

Upward trend with strong growth after 2023

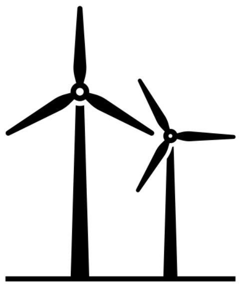
Strong annual seasonality (cycle 8760 h)

Hetero-scedasticity of residuals



Solution - model selection

Using modelling approaches that are specifically designed to address the non-stationarity of data in both data sets



Models

Baseline:

- Holt-Winters
- SARIMAX
- Ensemble model (Polynomial plus Random Forest Regressor)

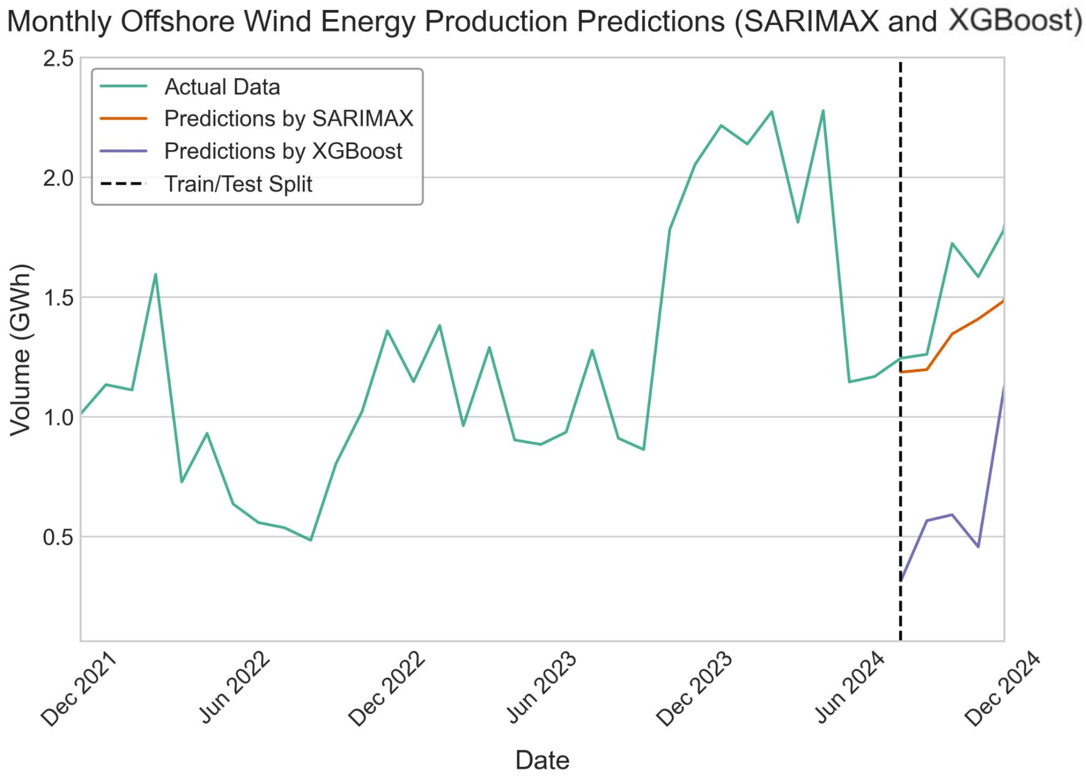
Complex:

- XGBOOST
- Chronos
- Prophet

Models' Performance (RMSE) — Offshore Energy Production				
Model	**Monthly**	**Weekly**	**Daily**	**Hourly**
SARIMAX	259 426	681 069	1 118 228	-
Ensemble	349 879	853 631	1 294 038	1 537 748
Holt-Winters	516 396	852 766	1 730 392	2 330 390
XGBoost	893 069	1 205 043	1 368 290	843 568
Prophet	343 193	861 173	1 252 612	-
Chronos	644 173	696 223	-	-

Models' Performance (RMSE) — Onshore Energy Production				
Model	**Monthly**	**Weekly**	**Daily**	**Hourly**
SARIMAX	175 868	819 331	1 258 253	-
Ensemble	414 909	971 487	1 461 657	1 625 782
Holt-Winters	774 005	974 837	2 106 683	2 527 093
XGBoost	983 691	1 258 568	1 451 051	891 734
Prophet	383 068	983 646	1 359 270	-
Chronos	469 942	750 766	-	-

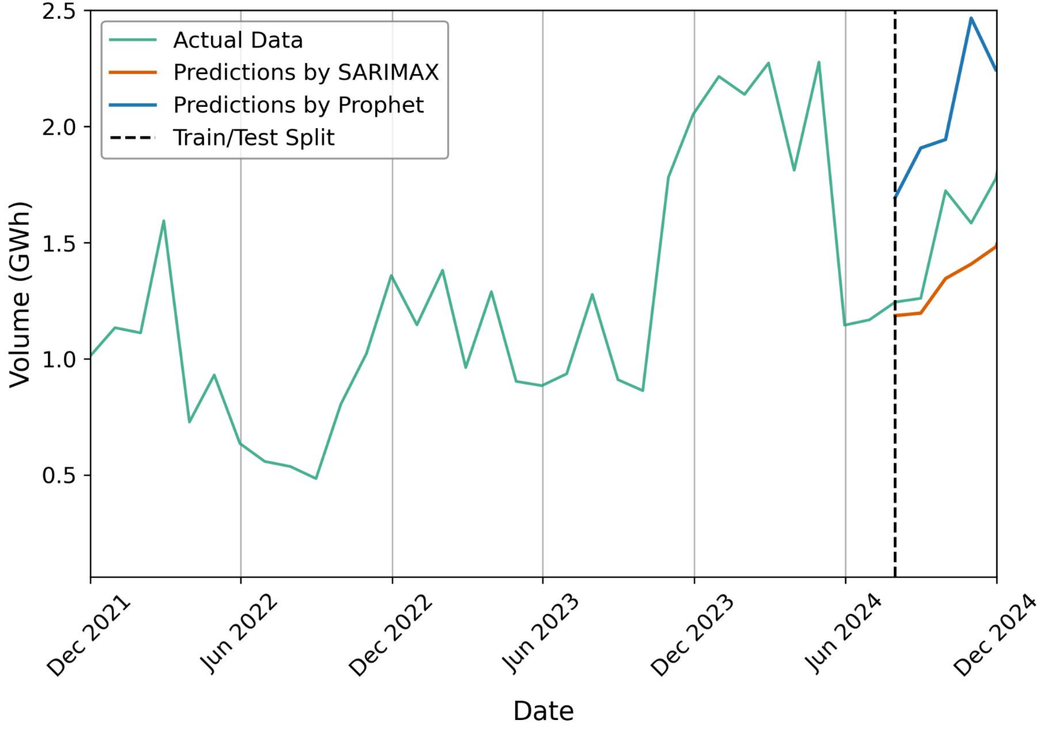
Results for MONTHLY data



- XGBOOST underestimates majorly
- Sarimax follows trend but not spikes in actual generation

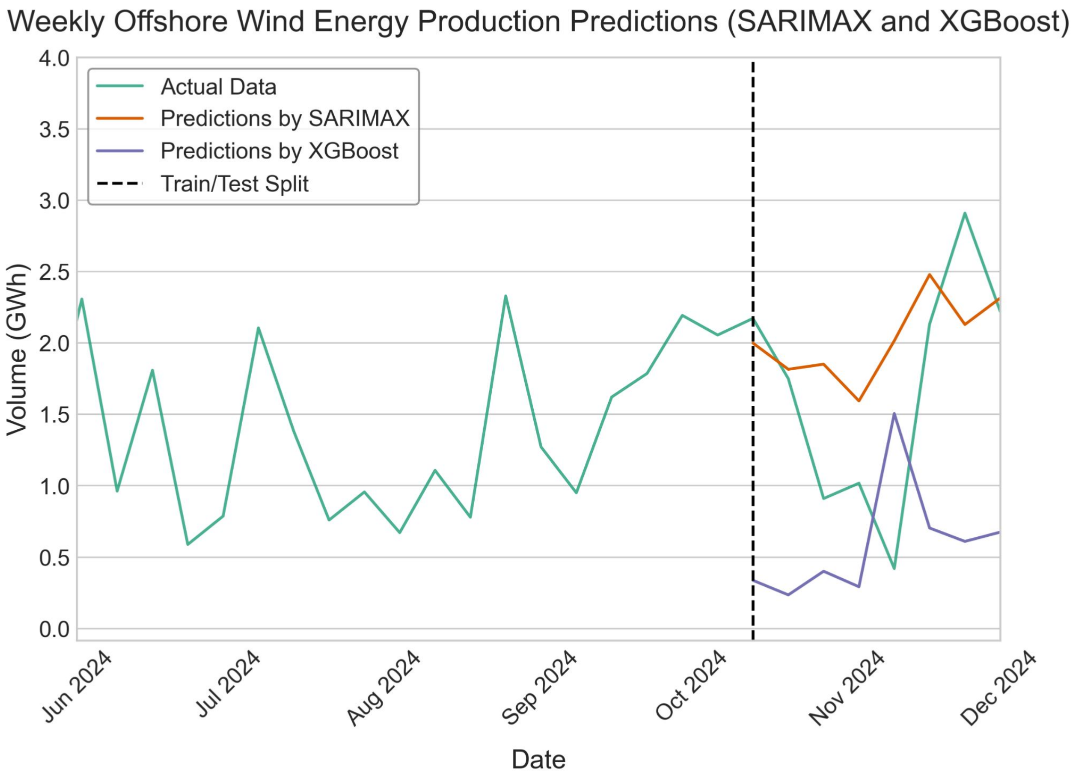
Results for MONTHLY data

Monthly Offshore Wind Energy Production Predictions (SARIMAX and Prophet)



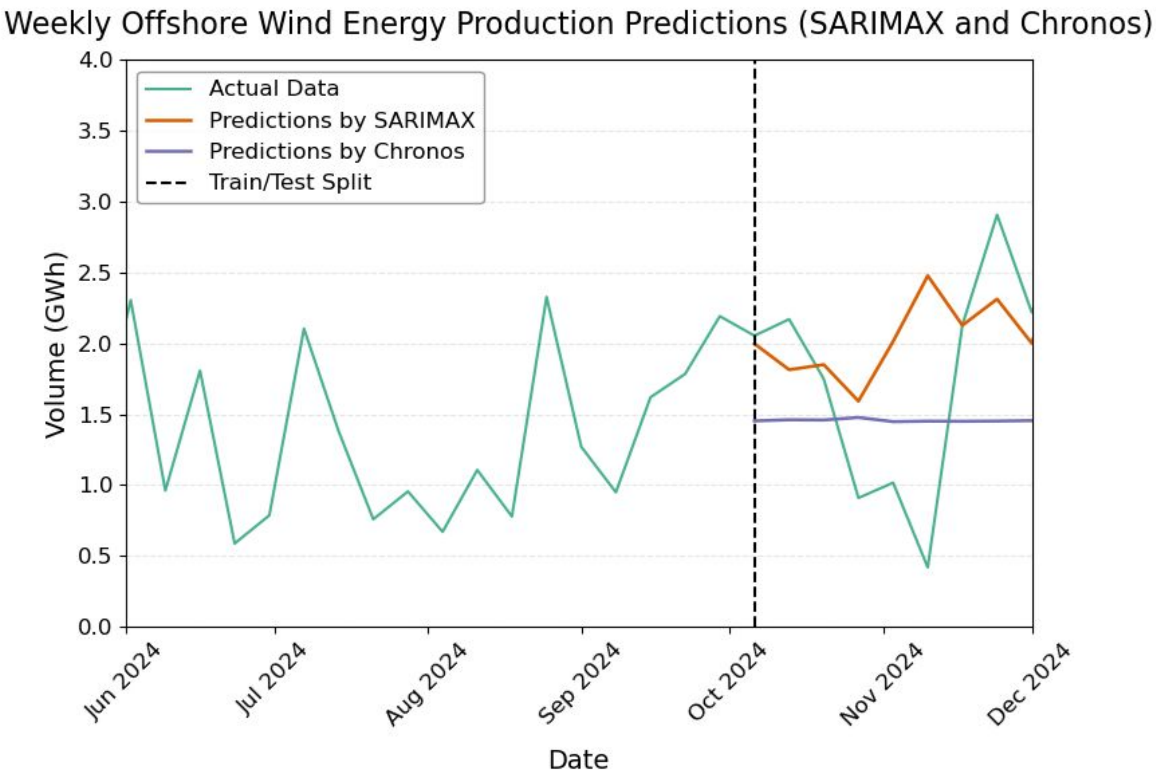
- Prophet overestimates majorly
- Sarimax follows trend but not spikes in actual generation

Results for WEEKLY data



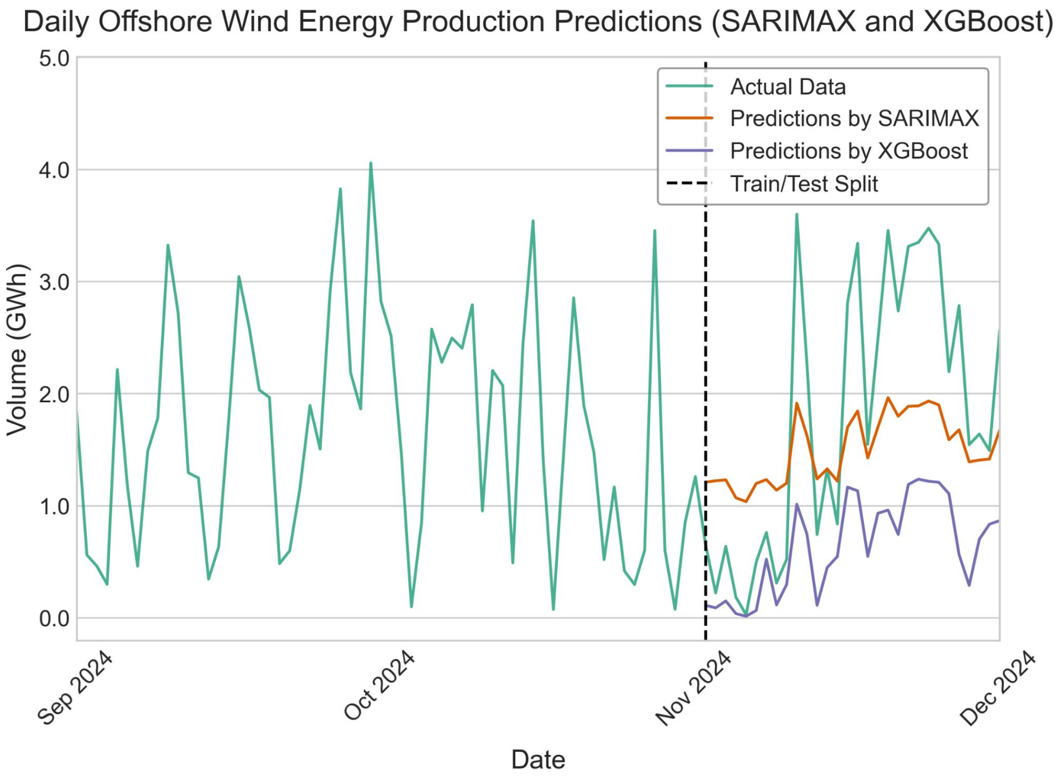
- Sarimax captures overall trend better
- XGBOOST still underestimates
- Both have lag in prediction

Results for WEEKLY data



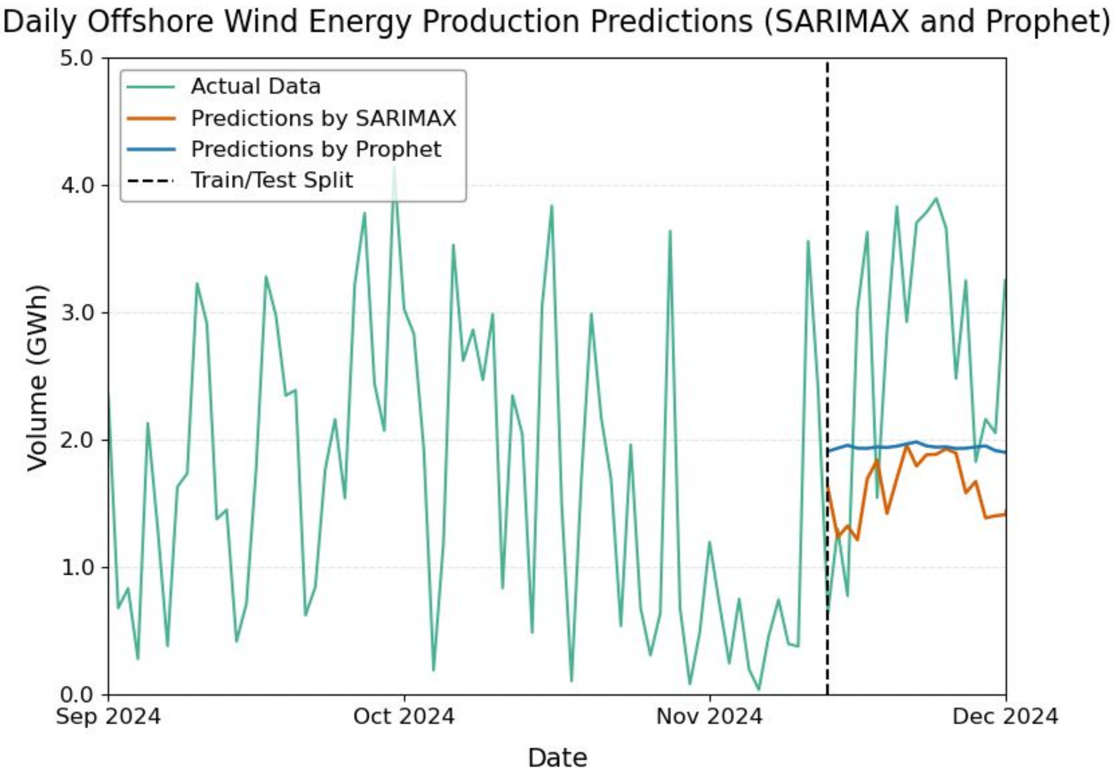
- Sarimax captures overall trend better
- Chronos does not capture the down- or upward trend

Results for DAILY data



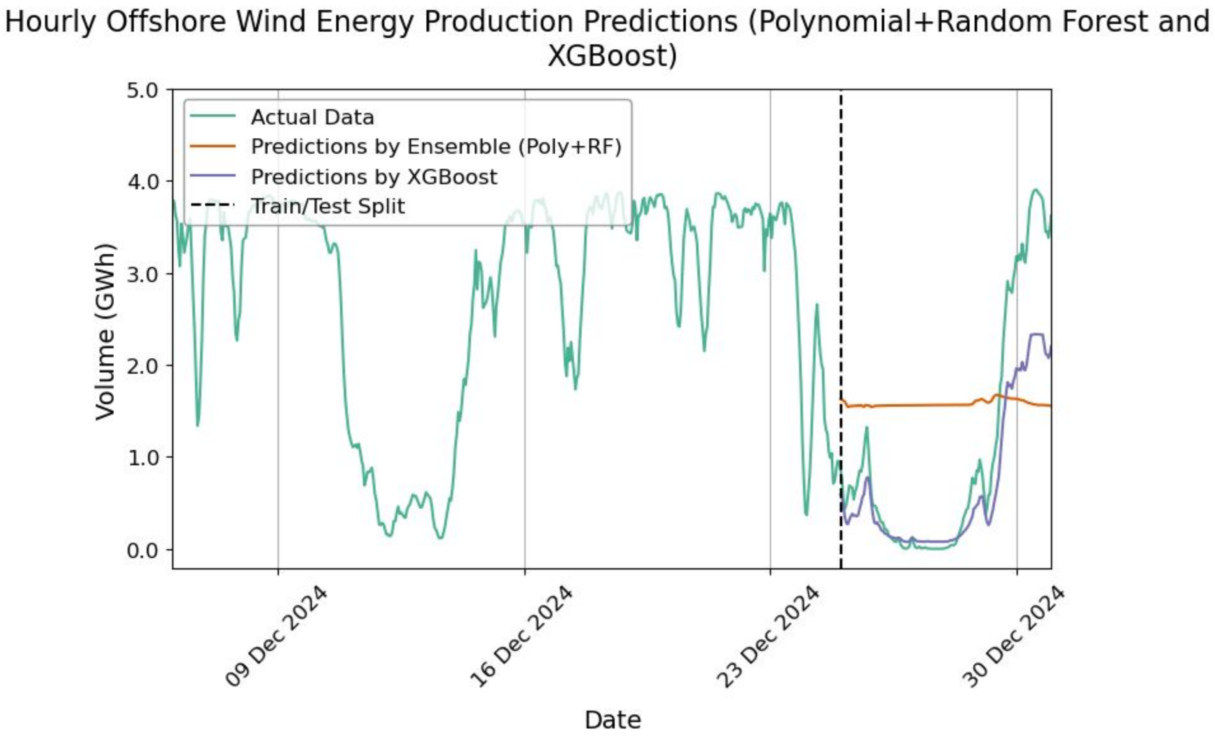
- Sarimax simultaneous spikes as actual data
- XGBOOST similar but again predicts lower generation

Results for DAILY data



- Sarimax simultaneous spikes as actual data
- Prophet stable predictions does not capture the down- or upward trend

Results for HOURLY data



- Ensemble (Poly+RF) stable predictions does not capture the down- or upward trend
- XGBOOST shows strength in difficult and large dataset, but overfits and underpredicts spike

Reflections and Next Steps

- Rolling time window to acknowledge temporal dimension better
- Difference in volume, not actual volume as label to combat non-stationarity
- Self-assigned weather stations to offshore and onshore not accurate enough
- Better market indicators (e.g. day-ahead and intra-day prices) instead of dynamic consumer prices
- Comparison with data from other countries
- Some models (LSTM, neural networks-based) require 3D-data: one dimension for individual units of analysis (turbines), one for time and one for target+exogenous variables, but such data are not publicly available
- Spatio-temporal models e.g. ConvLSTM on 5D-data

THANK YOU!

QUESTIONS?